Inpatient Aggression Prediction

8-Week Execution Plan

Contents

Inpatient Aggression Prediction — 8-Week Execution Plan

Consolidates and corrects `Predictive_Modeling`, `Violent_incidents`, and `Treatment_data`.

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0. Scope

Deliverable at week 8: a defensible retrospective feasibility study — a calibrated patient-shift aggression risk model, validated without leakage, benchmarked against the clinical incumbent, audited for subgroup performance, with an honest decision memo on whether it should proceed to deployment.

Not in scope at week 8: deployment, prospective validation, real-time scoring. Those are calendar-bound (IRB, IT integration, clinical governance), not effort-bound. No amount of hours compresses them.

The binding constraint is data access latency, not learning time. 12 hr/day for 8 weeks is ~600 hours, which is more than enough to reach working competence in every method below and execute the analysis. If extract access slips past week 2, the deliverable degrades to a design spec plus a power analysis — still presentable, but not a model.

Start the data request today, before reading anything else.

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1. Decision gates

Run these in order. Each one can kill or reshape the project. Do not build past a failed gate.

Gate	Week	Test	Fail action
G1 — Data exists	1	Incident reports timestamped and linkable to patient + ward + shift; staffing rosters at shift grain	Reframe as data-infrastructure proposal
G2 — Event count	2	≥ 200 positive shift-events, ideally ≥ 500	Aggregate to ward-shift; drop patient-level features
G3 — Ascertainment	3	Reporting rate not strongly confounded with staffing (see §3.2)	Switch target to mandatory-report classes only
G4 — Signal	5	Model beats base rate and beats DASA/HCR-20 if available	Report negative result — this is a legitimate finding
G5 — Calibration & fairness	7	Calibration slope near 1; no severe subgroup degradation	Do not recommend deployment

Events-per-variable rule of thumb: you need roughly 10–20 positive events per candidate predictor for a stable regression model. 300 events ≈ 15–30 predictors, maximum. Count your events before you count your features.

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2. Concept Reference (corrected)

2.1 Corrections to the original glossary

- **ROC-AUC** — measures ranking ability across all thresholds. **Not your headline metric.** At a <1% base rate it will read 0.85+ on a model that flags 40 patients to catch one incident. Report **PR-AUC (average precision)** as primary, ROC-AUC as secondary.
- **Random Forest** — bagging *plus random feature subsetting at each split*. The feature subsetting is what decorrelates the trees; without it you have bagged trees, not a forest.
- **Logistic regression** — the logit link maps probability → log-odds. The *inverse* logit (sigmoid) maps the linear predictor → probability.
- **Gradient boosting** — each tree fits the negative gradient of the loss (pseudo-residuals). That equals the raw residual only under squared error.

2.2 Additions — the concepts the glossary omitted

Three-way split / nested CV. Train / validation / test. Tune on validation, touch test once. Tuning against the test set is overfitting to the test set, and it is the most common way competent analysts produce optimistic numbers.

Calibration. Distinct from discrimination. A model can rank perfectly and still say "30%" when the true rate is 5%. You are producing a number that clinicians and executives will read as a probability, so calibration is the deliverable, not a nicety. - Measure: Brier score, calibration curve, calibration slope and intercept. - Fix: Platt scaling or isotonic regression, fit on the validation fold. - Warning: **any resampling for class imbalance (SMOTE, undersampling) destroys calibration.** Use class weights or Firth penalization instead.

Bias–variance tradeoff. The spine underneath overfitting, regularization, and ensembling. High bias = underfit, model too rigid. High variance = overfit, model too sensitive to the training sample. Regularization and bagging trade variance for bias.

Leakage. Any information in your features that would not have been available at the moment of prediction. The single most common cause of models that validate beautifully and fail completely. Treated in full in §3.3 because it is the primary threat to this specific project.

Interpretability. SHAP values (per-prediction feature attribution), permutation importance (global), partial dependence plots (marginal effect shape). In a state facility, "why was this patient flagged" will be asked by clinicians, by advocates, and possibly under oath. A model you cannot explain per-case is not deployable here regardless of its metrics.

Subgroup / fairness auditing. Compute PR-AUC, calibration slope, and false-positive rate at the operating threshold, separately by race, ethnicity, primary diagnosis, commitment type, and age band. Differential false-positive rates in a forensic setting are a civil-rights exposure, not just an ethics checkbox. Report them whether or not anyone asks.

Threshold selection. A probability is not a decision. Choose the operating threshold from the cost ratio of false positives to false negatives, and validate with **decision curve analysis** (net benefit across threshold probabilities) rather than by maximizing F1. F1 weights precision and recall equally, which is almost never the correct operational tradeoff.

Missing data. Distinguish missing-completely-at-random, missing-at-random, and missing-not-at-random. In clinical data, missingness is usually informative — an unrecorded assessment means something. Add missingness indicators; use multiple imputation only where justified. Never mean-impute a clinical variable silently.

Discrete-time survival framing. A viable alternative to per-shift binary: model the hazard of a first/next incident per shift with a complementary log-log link, which handles patient entry/exit and varying exposure windows naturally. Worth one afternoon of evaluation in week 3 — if patients have highly variable lengths of stay, this framing is cleaner.

2.3 Formula Reference

Notation: n observations, $y_i \in \{0, 1\}$ the observed outcome, $\hat{p}_i$ the predicted probability, $\mathbf{x}_i$ the feature vector, β the coefficient vector.

Ordinary least squares loss

$$\min_{\beta} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Logistic regression. The logit link maps probability to the linear predictor; the inverse logit maps back:

$$\operatorname{logit}(p_i) = \log \frac{p_i}{1 - p_i} = \mathbf{x}_i^\top \beta \qquad \Longleftrightarrow \qquad p_i = \sigma\left(\mathbf{x}_i^\top \beta\right) = \frac{1}{1 + e^{-\mathbf{x}_i^\top \beta}}$$

Binary cross-entropy (log loss)

$$J(\beta) = -\frac{1}{n} \sum_{i=1}^{n} \left[\, y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \,\right]$$

Elastic net penalty. $\alpha = 1$ gives lasso, $\alpha = 0$ gives ridge.

$$\min_{\beta} \; -\ell(\beta) + \lambda \left[\alpha \|\beta\|_1 + \tfrac{1-\alpha}{2} \|\beta\|_2^2\right]$$

Firth penalized likelihood. The Jeffreys-prior penalty on the Fisher information $\mathcal{I}(\beta)$ removes small-sample bias and resolves complete separation. This is the pre-planned fallback when the GLMM fails to converge.

$$\ell^*(\beta) = \ell(\beta) + \tfrac{1}{2}\log|\mathcal{I}(\beta)|$$

Mixed-effects logistic (GLMM). Shift i, patient k, ward j:

$$\operatorname{logit}(P(y_{ijk}=1)) = \mathbf{x}_{ijk}^\top\beta + u_j + v_{jk}, \qquad u_j \sim \mathcal{N}(0,\sigma^2_{\text{ward}}),\;\; v_{jk} \sim \mathcal{N}(0,\sigma^2_{\text{pat}})$$

Use crossed rather than nested random effects if patients transfer between wards.

Discrete-time hazard (alternative framing). h_{it} is the probability of a first incident in shift t given none through $t-1$; α_t is a flexible baseline in shift index.

$$\operatorname{cloglog}(h_{it}) = \log(-\log(1-h_{it})) = \alpha_t + \mathbf{x}_{it}^\top\beta$$

Precision, recall, F1

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}, \qquad F_1 = 2\cdot\frac{\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}}$$

F_1 weights precision and recall equally. Use it only if that is genuinely the operational tradeoff — here it almost certainly is not.

Average precision (PR-AUC) — the primary metric. Summed over thresholds n:

$$\text{AP} = \sum_n (R_n - R_{n-1})\,P_n$$

Brier score — calibration and discrimination combined. Lower is better.

$$\text{BS} = \frac{1}{n}\sum_{i=1}^{n}(\hat{p}_i - y_i)^2$$

Calibration slope and intercept. Refit the outcome on the logit of the predictions:

$$\operatorname{logit}(y_i) = a + b\cdot\operatorname{logit}(\hat{p}_i)$$

Perfect calibration is $a=0$, $b=1$. A slope $b<1$ means predictions are too extreme — the standard signature of overfitting. Report both, always, with the PR-AUC.

Net benefit (decision curve analysis). At threshold probability p_t, the odds $p_t/(1-p_t)$ is the exchange rate between a false positive and a true positive:

$$\text{NB}(p_t) = \frac{\text{TP}}{n} - \frac{\text{FP}}{n}\cdot\frac{p_t}{1-p_t}$$

Compare the model against treat-all and treat-none across the clinically plausible range of p_t. This is how you defend the operating threshold, not by maximizing F_1.

Root mean squared error (regression outcomes only — severity scores, not the binary target):

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Events per variable

$$\text{EPV} = \frac{\text{number of positive events}}{\text{number of candidate predictors}} \;\geq\; 10$$

2.4 The base-rate arithmetic — put this in the executive memo

Positive predictive value follows from Bayes' theorem, with prevalence π, sensitivity Se, and specificity Sp:

$$\mathrm{PPV} = \frac{\mathrm{Se} \cdot \pi}{\mathrm{Se} \cdot \pi + (1 - \mathrm{Sp})(1 - \pi)}$$

Take a genuinely good model — Se = 0.80, Sp = 0.90, which would post a ROC-AUC near 0.90 — against a per-patient-shift base rate of $\pi = 0.005$:

$$\mathrm{PPV} = \frac{0.80 \times 0.005}{0.80 \times 0.005 + 0.10 \times 0.995} = \frac{0.0040}{0.1035} \approx 0.039$$

Roughly 26 flagged shifts for every real incident. The model is not broken; this is what a strong classifier looks like against a rare event. Lead the executive memo with this number rather than the AUC, because someone else will compute it eventually and it is far better coming from you. It is also the number that determines whether the intervention is affordable — 26 extra checks per incident may be entirely reasonable for a nursing round and entirely unreasonable for anything costlier.

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3. Study Design Specification

3.1 Unit of analysis and target

Unit: patient-shift. One row per patient per shift they were present on the unit.

Target: binary occurrence of physical aggression toward others during the shift.

Corrections to the original spec:

- **Decide severity deliberately.** "Any physical aggression" and "injury-causing aggression" are different outcomes with different predictors and very different base rates. Score severity with **SOAS-R** if incident narratives support it, and pre-specify a primary outcome. Running both and reporting the better one is a specification-search failure that a reviewer will catch.
- **Separate outcome classes.** Verbal aggression, physical aggression toward others, self-directed harm, sexual aggression, and property destruction have distinct predictor profiles. Do not collapse them into a single "incident" flag.
- **Handle patient entry and exit.** Admissions and discharges mid-window. Length of stay is simultaneously a feature and an exposure window; if you treat it only as a feature you will confound duration with risk.

3.2 Ascertainment bias — address before modeling

Incident reports are not incidents. Reporting propensity is a function of staffing level, shift culture, and staff tenure — which are your primary predictors. A fully staffed shift with experienced staff documents more thoroughly. Left unaddressed, the model learns *more staff → more violence*, and that finding will be both wrong and career-damaging when presented.

Do all three:

1. **Triangulate the outcome.** Cross-check incident-report rates against less discretionary records over the same shifts: restraint and seclusion orders, injury/first-aid logs, stat medication administration, ER transfers. If incident reports track staffing but these do not, you have confirmed the artifact.
2. **Model reporting propensity explicitly.** Regress report count per shift on staffing and tenure with no patient covariates. A strong relationship is your bias estimate, and belongs in the limitations section either way.

3. **Have a fallback target.** Restrict to incident classes with mandatory reporting triggers (any restraint use, any injury, any law-enforcement notification) where discretion is minimal. Lower N, cleaner signal.

Document the result of this analysis regardless of outcome. It is independently publishable and it is what distinguishes this from a naïve dashboard exercise.

3.3 Timestamp discipline — the leakage protocol

Every feature gets a **hard availability timestamp**. The prediction is made at shift start time T. Any feature whose value was recorded, entered, or knowable only after T is excluded.

Build the feature matrix as an **as-of join** against T, not as a shift-level aggregate.

Specific traps in this dataset:

- **PRN medications.** Frequently administered *in response* to the escalation that becomes the incident. Same-shift PRN is leakage. Use prior-shift PRN counts only.
- **Progress notes.** A note authored at 1900 describing the 1830 incident is the outcome, not a predictor. Filter by *authored* timestamp, not by service date — these differ, often by hours.
- **Risk assessment scores.** Use the most recent score with a timestamp strictly before T. Scores are frequently updated *because* an incident occurred.
- **Ward census and staffing.** Use the assigned/scheduled roster known at shift start, not the actual worked roster reconciled afterward. If only reconciled data exists, say so.
- **Ward-level agitation trends.** Strictly lagged windows only: prior shift, prior 24h, prior 7 days, all ending before T.

3.4 Features

Environmental (shift grain): staff-to-patient ratio, staff tenure mix, registry/float staff proportion, ward census, admissions and discharges in prior 24h, time-of-day, day-of-week, holiday flag, ward identity.

Patient (as-of T): most recent structured risk score (DASA-IV / HCR-20 V3 / START), length of stay to date, primary and secondary psychiatric diagnosis category, commitment type, age, prior-incident counts over lagged windows, prior-shift PRN count, prior seclusion/restraint history.

Lagged behavioral: incidents in prior shift, prior 24h, prior 7d, at both patient and ward level. This captures behavioral momentum and ward contagion, and is usually the strongest predictor block.

Effective sample size warning. Patient-shift rows inflate N, but environmental features vary only at the shift level. Your statistical power for staffing effects is the **number of shifts**, not the number of rows. A dataset with 200,000 rows and 1,100 shifts has 1,100 observations of staffing. Compute both numbers and report both.

3.5 Modeling sequence

1. **Base rate.** Report it. Every subsequent metric is judged against it.
2. **Incumbent comparator.** If DASA-IV or HCR-20 scores exist in the record, compute their PR-AUC on your data. **This is the number to beat.** If no structured tool is in use, document that — implementing DASA-IV is a cheaper, faster win than building a model, and saying so is a stronger position than pretending otherwise.
3. **Regularized logistic regression** (elastic net) with cluster-robust standard errors. Interpretable baseline. Class weights, not resampling.
4. **Mixed-effects logistic (GLMM)** with random intercepts for patient nested in ward.
    - **Expect convergence failure.** Rare binary outcomes with crossed random effects frequently fail to converge or hit complete separation.

  - **Fallback, pre-planned:** Firth-penalized logistic regression with cluster-robust SEs. Decide now that you will use it, so you are not improvising in week 5.
5. **Gradient boosting** (LightGBM or XGBoost) to capture non-linear interactions between baseline patient risk and environmental load. Monotonic constraints where clinically sensible.
6. **Calibrate** the selected model on the validation fold. Re-check after calibration.
7. **SHAP** on the final model. Sanity-check the top features against clinical plausibility. If ward identity or staff tenure dominates, revisit §3.2.

3.6 Validation

- **Grouped CV by patient** — no patient appears in both train and test. Non-negotiable.
- **Rolling-origin (forward-chaining) CV by time** — train on earlier periods, test on later. Catches temporal drift and policy changes.
- **Leave-one-ward-out** — tests whether the model generalizes across units or has memorized ward-specific patterns.
- **Test set touched once.** Lock it in week 3.

Use all three schemes and report all three. Divergence between them is informative, not a problem to hide.

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4. Text Features — Scoped to Fit

The original NLP plan is a year of work. Compressed to what returns value in 8 weeks:

In scope (~1 week, weeks 5–6):

- **Lexicon/regex counts** over prior-shift notes only. Behavioral markers: pacing, refusal, verbal escalation, threat language, intrusiveness, medication refusal, sleep disruption. Highest value per hour of any text method here.
- **Negation and assertion handling.** Mandatory, not optional. “Denies auditory hallucinations” scores as a positive hit without it, which inverts the feature’s meaning. Use `medspaCy` (ConText algorithm) or `negspacy`.
- **Copy-forward deduplication.** EHR templates and carried-forward text inflate every count. Hash-dedupe repeated blocks before counting.
- **Timestamp filtering.** Per §3.3 — authored timestamp strictly before T. This is the difference between a real feature and a leak.
- **Lexicon validation.** Hand-annotate 200–300 note segments, compute precision and recall of the lexicon against them. Without this you cannot claim the feature measures what you say.

Deferred:

- **ClinicalBERT embeddings.** MIMIC-III (ICU) pretrained; weak transfer to forensic psychiatric nursing narratives, 512-token cap, and 768 dense dimensions against a few hundred positive events is a dimensionality disaster. Also requires on-prem GPU. If text features prove valuable via the lexicon route, this becomes a justified phase-2 ask.
- **LDA topic modeling.** Unstable on short clinical notes, rarely yields actionable topics. Cut it.

Sequencing rule: the tabular model must have a stable PR-AUC *before* text features are added, or you cannot attribute any lift to them. Text is an ablation study, not a parallel track.

Compliance: de-identification and secure processing are not alternatives. Regex de-ID of clinical free text does not meet HIPAA Safe Harbor. All processing on-prem, inside the approved environment, no external API calls, full stop.

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5. Eight Weeks

Assumes ~70 hr/week. Learning and execution are interleaved deliberately — you learn each method in the week you need it, not in a front-loaded theory block.

Week 1 — Access and foundations Submit the data extract request and the governance packet on day 1. Concurrently: logistic regression, regularization, train/validation/test discipline, PR-AUC vs ROC-AUC, calibration. Set up the environment and version control. Write the pre-analysis plan draft (§6).

Week 2 — Data assembly and G2 Build the patient-shift spine. Assemble incident reports, rosters, census, risk scores. Count events. Compute events-per-variable. Learn: mixed models conceptually, hierarchical data structure.

Week 3 — Ascertainment audit and G3 Execute §3.2 in full. Build the as-of feature join with hard timestamps (§3.3). Lock the test set. Evaluate the discrete-time survival framing as an alternative. Learn: leakage patterns, survival basics.

Week 4 — Baseline models Base rate, incumbent comparator, elastic-net logistic. First honest PR-AUC. Learn: GLMM fitting, Firth penalization, class weighting.

Week 5 — GLMM and boosting, G4 Mixed-effects model with the Firth fallback ready. LightGBM benchmark. Grouped and rolling-origin CV. Learn: gradient boosting mechanics, hyperparameter tuning within CV.

Week 6 — Text features Lexicon, negation, dedup, validation, ablation against week 5 results. Learn: clinical NLP basics only — resist scope creep here.

Week 7 — Calibration, interpretation, fairness, G5 Calibrate. SHAP and permutation importance. Full subgroup audit. Decision curve analysis and threshold selection. Learn: calibration methods, DCA, fairness metrics.

Week 8 — Write and defend Technical report, executive memo, limitations section, deployment recommendation with governance requirements. Build the "what would change my mind" section — pre-empt the objections rather than absorbing them live.

Slippage protocol: if data access lands in week 3 or later, drop the text work entirely and shift everything right by the delay. If it lands week 5 or later, the deliverable becomes the design spec, the ascertainment analysis on whatever data exists, and a power analysis. Say so early rather than presenting a rushed model.

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6. Governance

Start these in week 1. They run on institutional time, not yours.

- ☐ Data use request through DSH research/IT channels, with the exact field list
- ☐ IRB / research committee determination — even for retrospective QI work, get the determination in writing
- ☐ HIPAA minimum-necessary justification for each requested field
- ☐ **Pre-analysis plan, timestamped and filed before touching the test set** — primary outcome, primary metric, comparator, subgroup list, decision thresholds. This is both methodological hygiene and self-protection.
- ☐ Written scope statement: this model informs staffing and clinical attention only. Any use touching seclusion, restraint, privilege level, or release recommendation is a different legal category requiring separate review. Put the boundary in writing before anyone else defines it for you.

☐ Post-deployment drift monitoring plan (specified now, executed later)

Performativity note for the memo: if the model's output changes staffing, the intervention alters the outcome the model predicts, and measured performance will degrade after deployment. This is expected behavior, not model failure. Stating it up front prevents it being read as one later.

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7. Deliverables at Week 8

1. **Technical report** — design, ascertainment analysis, model comparison, validation across three CV schemes, calibration, subgroup audit, limitations.
2. **Executive memo, 2 pages** — what it predicts, how well, versus what alternative, what it would cost to deploy, what it must not be used for.
3. **Reproducible pipeline** — versioned, documented, runnable by someone else.
4. **Pre-analysis plan** — filed before analysis, referenced in the report.
5. **Recommendation** — proceed, proceed with conditions, or do not proceed. A well-evidenced "do not proceed" is a successful project outcome and should be framed that way from day one.

Appendix — Stack

R is stronger for the mixed-model work; **Python** is stronger for boosting, SHAP, and text. Use both. Neither requires GPU.

Task	R	Python
GLMM	`lme4::glmer`, `glmmTMB`	`pymer4`, `statsmodels` (limited)
Firth penalized	`logistf`	`firthlogist`
Elastic net	`glmnet`	`sklearn` `LogisticRegression(penalty='elasticnet')`
Boosting	`lightgbm`	`lightgbm`, `xgboost`
Grouped/temporal CV	`rsample`	`StratifiedGroupKFold`, `TimeSeriesSplit`
Calibration	`rms::val.prob`	`sklearn.calibration` (`CalibratedClassifierCV`, `calibration_curve`)
PR-AUC	`PRROC`	`sklearn.metrics.average_precision_score`
Interpretation	`DALEX`	`shap`
Decision curves	`dcurves`	`dcurves`
Clinical negation	—	`medspacy`, `negspacy`

Minimum viable ask if access is contested: read access to the incident, roster, census, and assessment tables, plus Python or R on a workstation inside the secure environment. No GPU, no external network, no PHI egress. That is a defensible, narrow request — and it is the one that produces a demonstrable result.